

Bundle Structures in Topology

Labix

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Abstract

- Notes on Algebraic Topology by Oscar Randal-Williams

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1 Fibrations and Cofibrations

1.1 Fibrations and The Homotopy Lifting Property

Definition 1.1.1 (The Homotopy Lifting Property) Let $p : E \rightarrow B$ be a map and let X be a space. We say that p has the homotopy lifting property with respect to X if for every homotopy $H : X \times I \rightarrow B$ and a lift $\widetilde{H(-, 0)} : X \rightarrow E$ of $H(-, 0)$, there exists a homotopy $\tilde{H} : X \times I \rightarrow E$ such that the following diagram commutes:

$$\begin{array}{ccc} X \times \{0\} & \xrightarrow{\widetilde{H(-, 0)}} & E \\ \downarrow \iota & \nearrow \exists \tilde{H} & \downarrow p \\ X \times I & \xrightarrow{H} & B \end{array}$$

Definition 1.1.2 (Fibrations) We say that a map $p : E \rightarrow B$ is a fibration if it has the homotopy lifting property with respect to all topological spaces X . We call B the base space and E the total space.

Definition 1.1.3 (The Hopf Fibration) Define the Hopf fibration $h : S^3 \rightarrow S^2$ as follows. Consider S^2 as the one point compactification of $\mathbb{C}$. Also consider $S^3 = \{(z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^2 + |z_2|^2 = 1\}$. Define the map h by

$$(z_1, z_2) \rightarrow \frac{z_2}{z_1}$$

Example 1.1.4 The Hopf fibration $h : S^3 \rightarrow S^2$ is a fibration. Moreover, the fibers of the Hopf fibration are circles S^1 .

Proof We can rewrite the coordinates of S^3 by $r_j e^{i\theta_j}$. Then

$$h(r_1 e^{i\theta_1}, r_2 e^{i\theta_2}) = \frac{r_2}{r_1} e^{i(\theta_2 - \theta_1)}$$

Fix $r e^{i\theta} \in S^2$. Then there exists a unique pair (r_1, r_2) that solves the simultaneous equation $rr_1 = r_2$ and $r_1^2 + r_2^2 = 1$. ■

1.2 Cofibrations and The Homotopy Extension Property

Definition 1.2.1 (The Homotopy Extension Property) Let $i : A \rightarrow X$ be a map and let Y be a space. Denote i_0 the inclusion map $A \times \{0\} \hookrightarrow A \times I$. We say that i has the homotopy extension property with respect to Y if for every homotopy $H : A \times I \rightarrow Y$ and every map $f : X \rightarrow Y$ such that

$$H \circ i_0 = f \circ i$$

there exists a homotopy $\tilde{H} : X \times I \rightarrow Y$ such that the following diagram commute:

$$\begin{array}{ccc} A \cong A \times \{0\} & \xrightarrow{\iota_0} & A \times I \\ \downarrow i & & \downarrow i \times \text{id}_I \\ X \cong X \times \{0\} & \xrightarrow{\iota_0} & X \times I \end{array} \quad \begin{array}{c} \nearrow H \\ \searrow \exists \tilde{H} \\ \downarrow f \end{array}$$

Definition 1.2.2 (Cofibrations) Let A, X be spaces. Let $i : A \rightarrow X$ be a map. We say that i is a cofibration if it has the homotopy extension property for all spaces Y .

Proposition 1.2.3 Let A, X be spaces. Let $i : A \rightarrow X$ be a cofibration. Then $i : A \rightarrow i(A)$ is a homeomorphism.

There is actually an easier way to write out cofibrations when (X, A) is a pair of spaces.

Lemma 1.2.4 Let (X, A) be a pair of spaces with A closed in X . Let $\iota : A \rightarrow X$ be the inclusion. Then ι is a cofibration if and only if for all spaces Y and maps $f : X \rightarrow Y$ and $H : A \times I \rightarrow Y$, there exists a map $\tilde{H} : X \times I \rightarrow Y$ such that the following diagram commutes:

$$\begin{array}{ccc} X \times \{0\} \cup A \times I & \xrightarrow{f \cup H} & Y \\ \downarrow \iota & \nearrow \tilde{H} & \\ X \times I & & \end{array}$$

1.3 Basic Properties of Fibrations and Cofibrations

Proposition 1.3.1 Let X_1, X_2, Y_1, Y_2 be spaces. Let $p_1 : X_1 \rightarrow Y_1$ and $p_2 : X_2 \rightarrow Y_2$ be maps. Then the following are true.

- If p_1 and p_2 are fibrations then $p_1 \times p_2 : X_1 \times X_2 \rightarrow Y_1 \times Y_2$ is a fibration.
- If p_1 and p_2 are cofibrations then $p_1 \amalg p_2 : X_1 \amalg X_2 \rightarrow Y_1 \amalg Y_2$ is a cofibration.

Proposition 1.3.2 Let X, Y, Z be spaces. Let $f : X \rightarrow Y$ be a map.

- Let f be a fibration. Consider the following lifting problem:

$$\begin{array}{ccc} Z \times \{0\} & \xrightarrow{g} & X \\ \downarrow i_0 & \nearrow & \downarrow f \\ Z \times I & \xrightarrow{h} & Y \end{array}$$

If h_0 and h_1 are both solutions to the lifting problem, then h_0 and h_1 are homotopic relative to $Z \times \{0\}$.

- Let f be a cofibration. Consider the following extension problem:

$$\begin{array}{ccc} X & \xrightarrow{g} & Z \times \{0\} \\ \downarrow f & \nearrow & \downarrow \text{ev}_0 \\ Y & \xrightarrow{h} & Z \times I \end{array}$$

If h_0 and h_1 are both solutions to the extension problem, then h_0 and h_1 are homotopic relative to Z .

1.4 Serre Fibrations

Definition 1.4.1 (Serre Fibration) We say that a map $p : E \rightarrow B$ is a Serre fibration if it has the homotopy lifting property with respect to all CW-complexes.

Lemma 1.4.2 Every (Hurewicz) fibration is a Serre fibration.

Proof This is true since Hurewicz fibrations satisfies the homotopy lifting property with respect to all topological spaces, including CW complexes. ■

2 Vector Bundles

2.1 Basic Definitions

Definition 2.1.1 (Vector Bundles) Let F be a field. A vector bundle (E, B, p) consists of two topological spaces E and B , a continuous surjection $p : E \rightarrow B$ such that

- For every $b \in B$, the fibre $E_b = p^{-1}(b)$ is an F -vector space of dimension k .
- For every $b \in B$, there exists an open neighbourhood $U \subseteq B$ of p and a homeomorphism $\phi : p^{-1}(U) \rightarrow U \times F^k$ such that for $\pi : U \times F^k \rightarrow U$ the projection map, the following diagram commutes

$$\begin{array}{ccc} p^{-1}(U) & \xrightarrow{\phi} & U \times F^k \\ & \searrow p & \swarrow \pi \\ & U & \end{array}$$

and the map

$$E_b \xrightarrow{\phi|_{E_b}} \{b\} \times F^k \xrightarrow{\pi} F^k$$

is a vector space isomorphism.

B is said to be the base space and E the total space. Each (U, ϕ) is said to be a local trivialization.

The local trivialization means that locally at a neighbourhood, the vector bundle looks the same the open set times F^k . In particular, there is also a notion of trivial bundle which means that the bundle is globally just $B \times \mathbb{R}^r$.

Definition 2.1.2 (Sections) A section of a vector bundle $p : E \rightarrow B$ is a map $s : B \rightarrow E$ assigning to each $b \in B$ a vector space $s(b)$ in the fiber $p^{-1}(b)$.

Proposition 2.1.3 Let $p : E \rightarrow B$ be a vector bundle. Let s, s_1, s_2 be sections of E . Then $s_1 + s_2$ and λs are also vector bundles for any $\lambda \in \mathbb{R}$. Moreover, the set of all sections $s(E)$ is a vector space.

Definition 2.1.4 (Morphism of Vector Bundles) Let $p_1 : E_1 \rightarrow B_1$ and $p_2 : E_2 \rightarrow B_2$ be vector bundles. A morphism of these vector bundles is given by a pair of continuous maps $f : E_1 \rightarrow E_2$ and $g : B_1 \rightarrow B_2$ such that the following diagram commutes

$$\begin{array}{ccc} E_1 & \xrightarrow{f} & E_2 \\ \downarrow p_1 & & \downarrow p_2 \\ B_1 & \xrightarrow{g} & B_2 \end{array}$$

If $B = B_1 = B_2$ then the diagram collapses:

$$\begin{array}{ccc} E_1 & \xrightarrow{f} & E_2 \\ & \searrow p_1 & \swarrow p_2 \\ & B & \end{array}$$

Definition 2.1.5 (Isomorphism of Vector Bundles) A bundle homomorphism from E_1 to E_2 is an isomorphism if there exists an inverse bundle homomorphism from E_2 to E_1 . In this case, we say that E_1 and E_2 are isomorphic.

2.2 The Cocycle Conditions

Given two charts (U_α, ϕ_α) and (U_β, ϕ_β) of a vector bundle,

$$\phi_\beta \circ \phi_\alpha^{-1} : (U_\alpha \cap U_\beta) \times F^k \rightarrow (U_\alpha \cap U_\beta) \times F^k$$

is a well defined function. In particular, by fixing a point in $U_\alpha \cap U_\beta$, we obtain a linear map.

Definition 2.2.1 (Transition Functions) Let $p : E \rightarrow B$ be an F -vector bundle of rank r . Let (U_α, ϕ_α) and (U_β, ϕ_β) be local trivialization. For each $x \in U_\alpha \cap U_\beta$, $\phi_\beta \circ \phi_\alpha^{-1}(x, -) : F^k \rightarrow F^k$ is a linear map. Define $g_{U_\alpha U_\beta} : U_\alpha \cap U_\beta \rightarrow \text{GL}(n, F)$ by

$$x \mapsto \phi_\beta \circ \phi_\alpha^{-1}(x, -) : F^k \rightarrow F^k$$

In other words, $g_{U_\alpha U_\beta}$ is such that

$$\phi_\beta \circ \phi_\alpha^{-1}(x, v) = (x, g_{U_\alpha U_\beta}(x)v)$$

For $x \in U_\alpha \cap U_\beta$ and $v \in F^k$.

Proposition 2.2.2 Let $p : E \rightarrow B$ be a K -vector bundle of rank r . The transition functions of the vector bundle satisfies the following.

- Cocycle condition: $g_{\alpha\beta} \circ g_{\beta\gamma} \circ g_{\gamma\alpha} = I_r$ on $U_\alpha \cap U_\beta \cap U_\gamma$
- $g_{\alpha\alpha} = I_r$ on U_α

2.3 Operations on Vector Bundles

Definition 2.3.1 (Whitney Sum) Let $p_1 : E_1 \rightarrow B$ and $p_2 : E_2 \rightarrow B$ be two vector bundles. Define the direct sum of the vector bundles to be

$$E_1 \oplus E_2 = \{(v_1, v_2) \in E_1 \times E_2 \mid p_1(v_1) = p_2(v_2)\}$$

together with the projection $p : E_1 \oplus E_2 \rightarrow B$ defined by $(v_1, v_2) \mapsto p_1(v) = p_2(v)$.

Lemma 2.3.2 The Whitney sum $E_1 \oplus E_2$ of two vector bundles is again a vector bundle.

Proposition 2.3.3 (Tensor Product Bundle) Let $p_1 : E_1 \rightarrow B$ and $p_2 : E_2 \rightarrow B$ be vector bundles. Define the tensor product bundle of it to be

$$E_1 \otimes E_2 = \{p_1^{-1}(x) \otimes p_2^{-1}(x) \mid x \in B\}$$

The construction $E_1 \otimes E_2$ is a vector bundle over B .

Theorem 2.3.4 (Pullback Bundle) Let $p : E \rightarrow Y$ be a vector bundle. Let $f : X \rightarrow Y$ be a continuous map. Then there exists E' and p' such that $p' : E' \rightarrow X$ is a vector bundle.

Theorem 2.3.5 (Dual Bundle) Let $p : E \rightarrow B$ be a K -vector bundle. Then the dual bundle $p^* : E^* \rightarrow B$ defined by

$$E_b^* = \text{Hom}_K(E_b, K)$$

is a vector bundle over B .

3 The Topology of Fiber Bundles

3.1 Fiber Bundles

Fiber bundles serve as somewhat of a generalization of both vector bundles and covering spaces, while being a special case of a fibration. It therefore has the properties of a fibration.

Definition 3.1.1 (Fiber Bundles) Let E, B, F be spaces with B connected, and $p : E \rightarrow B$ a continuous map. We say that p is a fiber bundle over F if the following are true.

- $p^{-1}(b) \cong F$ for all $b \in B$
- $p : E \rightarrow B$ is surjective
- Local Triviality: There exists a collection

$$\{(U_k \subseteq B, \phi_k : p^{-1}(U_k) \rightarrow U_k \times F) \mid k \in I\}$$

called local trivialization charts such that the following diagram commutes:

$$\begin{array}{ccc} p^{-1}(U_k) & \xrightarrow{\phi_k} & U_k \times F \\ & \searrow p & \swarrow \pi \\ & U_k & \end{array}$$

where π is the projection by forgetting the second variable.

We say that B is the base space, E the total space. It is denoted as (F, E, B)

Intuitively, we would like a fiber bundle to locally look like the product $B \times F$.

Fiber bundles generalizes vector bundles in the sense that the fibers are no longer vector spaces but instead arbitrary spaces.

Lemma 3.1.2 Every vector bundle is a fiber bundle.

Proof Indeed if $p : E \rightarrow B$ is a vector bundle, then each fiber $p^{-1}(b)$ is an n -dimensional vector spaces over a field F . Moreover, by definition the local triviality condition is also satisfied. ■

A lot of examples of fiber bundles therefore come from vector bundles. Another familiar collection of examples come from covering space theory.

Lemma 3.1.3 Every covering space is a fiber bundle.

Proof If $p : \tilde{X} \rightarrow X$ is a covering space, then we have seen that $p^{-1}(x)$ remains constant as $x \in X$ varies. Moreover, $p^{-1}(x)$ has the discrete topology with countable fiber since each $p^{-1}(U)$ is a disjoint union for $U \subseteq X$ open. Thus they must all be homeomorphic.

Finally, for any $U \subseteq X$, recall that

$$p^{-1}(U) = \coprod_{i \in I} V_i$$

where each $V_i \cong U$. It is clear by definition that $|p^{-1}(x)| = |I|$ for any $x \in X$. By giving I the discrete topology, we obtain a homeomorphism $p^{-1}(x) \cong I$. The homeomorphism $p^{-1}(U) = \coprod_{i \in I} V_i$ translates to

$$p^{-1}(U) = \coprod_{i \in I} V_i \cong \coprod_{i \in I} U \cong U \times I$$

defined by $\tilde{x} \in V_i \mapsto (p(\tilde{x}) = x, i)$. It is thus clear that the local triviality condition is satisfied. ■

Proposition 3.1.4 Every fiber bundle is a Serre fibration.

We can provide a partial converse for the fact that every fiber bundle is a Serre fibration.

Proposition 3.1.5 Let $p : E \rightarrow B$ be a fiber bundle. If B is paracompact, then p is a (Hurewicz) fibration.

We therefore have inclusions

$$\text{Fiber Bundles} \subset \text{Serre Fibrations} \subset \text{(Hurewicz) Fibrations}$$

Definition 3.1.6 (Map of Fiber Bundles) Let (F_1, E_1, B_1) and (F_2, E_2, B_2) be fiber bundles. A map of fiber bundles is a pair of basepoint preserving continuous maps $(\tilde{f} : E_1 \rightarrow E_2, f : B_1 \rightarrow B_2)$ such that the following diagram commutes:

$$\begin{array}{ccc} E_1 & \xrightarrow{\tilde{f}} & E_2 \\ p_1 \downarrow & & \downarrow p_2 \\ B_1 & \xrightarrow{f} & B_2 \end{array}$$

Such a map of fiber bundles determine a continuous of the fibers $F_1 \cong p_1^{-1}(b_1) \rightarrow p_2^{-1}(b_2) \cong F_2$.

A map of fiber bundles $(\tilde{f}, f)$ is said to be an isomorphism if there is a map $(\tilde{g} : E_2 \rightarrow E_1, g : B_2 \rightarrow B_1)$ such that $\tilde{g}$ is the inverse of $\tilde{f}$ and g is the inverse of f .

Notice that a morphism of fiber bundles preserves fibers. Indeed, If $p_1^{-1}(b)$ is a fiber of B , then using the commutativity of the diagram we have that

$$p_2(\tilde{f}(p_1^{-1}(b))) = f(p_1(p_1^{-1}(b))) = f(b)$$

which implies that

$$p_2^{-1}(f(b)) = \tilde{f}(p_1^{-1}(b))$$

or in other words, the fiber at $f(b)$ is the same as the fiber at b applied with $\tilde{f}$.

Definition 3.1.7 (Equivalent Fiber Bundles) Let $p : E_1 \rightarrow B_1$ and $p : E_2 \rightarrow B_2$ be two fiber bundles. We say that they are equivalent if there exists an isomorphism $(\tilde{f} : E_1 \rightarrow E_2, f : B_1 \rightarrow B_2)$ of fiber bundles.

There are two important special cases of fiber bundles that will appear time and time again.

Definition 3.1.8 (Trivial Bundles) We say that a fiber bundle (F, E, B) is trivial if (F, E, B) is isomorphic to the trivial fibration $B \times F \rightarrow B$.

Definition 3.1.9 (The Pullback Bundle) Let $p : E \rightarrow B$ be a fiber bundle with fiber F . Let $f : B' \rightarrow B$ be a continuous function. Define the pullback of p by f to be the space

$$f^*(E) = \{(b', e) \in B' \times E \mid p(e) = f(b')\}$$

Theorem 3.1.10 Let $p : E \rightarrow B$ be a fiber bundle. Suppose that $f, g : X \rightarrow B$ are homotopic maps. Then the pull back bundles

$$f^*(E) \cong g^*(E)$$

are equivalent.

3.2 Sections of a Bundle

Definition 3.2.1 (Sections) Let (F, E, B) be a fiber bundle. A section on the fiber bundle is a map $s : B \rightarrow E$ such that

$$p \circ s = \text{id}_B$$

Definition 3.2.2 (Local Sections) Let (F, E, B) be a fiber bundle. Let $U \subset B$ be an open set. A local section of the fiber bundle on U is a map $s : U \rightarrow E$ such that

$$p \circ s = \text{id}_U$$

3.3 Sphere Bundles

We now consider a special type of fibrations where the fibers are given by S^1 . When we pick $n = 1$ we obtain the classical object of study in algebraic topology called the Hopf fibration.

Definition 3.3.1 (Sphere Bundles) A sphere bundle is a fiber bundle $p : E \rightarrow B$ for which its fibers are the n -sphere S^n .

Theorem 3.3.2 Let $n \in \mathbb{N}$. Consider S^{2n+1} lying inside $\mathbb{C}^{n+1}$. Then canonical map $\mathbb{C}^n \rightarrow \mathbb{CP}^n$ given by

$$(z_0, \dots, z_n) \mapsto [z_0 : \dots : z_n]$$

is a fiber bundle with fiber S^1 .

Definition 3.3.3 (Hopf Fibration / Hopf Bundle) The fiber bundle $p : S^3 \rightarrow S^2$ with fiber S^1 is called the Hopf fibration / Hopf bundle.